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RESEARCH ARTICLE

An efficient memory reserving-and-fading strategy for vector quantization based 3D brain segmentation and tumor extraction using an unsupervised deep learning network

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Abstract

Deep learning networks are state-of-the-art approaches for 3D brain image segmentation, and the radiological characteristics extracted from tumors are of great significance for clinical diagnosis, treatment planning, and treatment outcome evaluation. However, two problems have been the hindering factors in brain image segmentation techniques. One is that deep learning networks require large amounts of manually annotated data. Another issue is the computational efficiency of 3D deep learning networks. In this study, we propose a vector quantization (VQ)-based 3D segmentation method that employs a novel unsupervised 3D deep embedding clustering (3D-DEC) network and an efficiency memory reserving-and-fading strategy. The VQ-based 3D-DEC network is trained on volume data in an unsupervised manner to avoid manual data annotation. The memory reserving-and-fading strategy beefs up model efficiency greatly. The designed methodology makes deep learning-based model feasible for biomedical image segmentation. The experiment is divided into two parts. First, we extensively evaluate the effectiveness and robustness of the proposed model on two authoritative MRI brain tumor databases (i.e., IBSR and BrainWeb). Second, we validate the model using real 3D brain tumor data collected from our institute for clinical practice significance. Results show that our method (without data manual annotation) has superior accuracy (0.74 ± 0.04 Tanimoto coefficient on IBSR, 97.5% TP and 97.7% TN on BrainWeb, and 91% Dice, 88% sensitivity and 87% specificity on real brain data) and remarkable efficiency (speedup ratio is 18.72 on IBSR, 31.16 on BrainWeb, 31.00 on real brain data) compared to the state-of-the-art methods. The results show that our proposed model can address the lacks of manual annotations, and greatly increase computation speedup with competitive segmentation accuracy compared to other state-of-the-art 3D CNN models. Moreover, the proposed model can be used for tumor treatment follow-ups every 6 months, providing critical details for surgical and postoperative treatment by correctly extracting numerical radiomic features of tumors.

Keywords 3D data segmentation · Unsupervised deep learning · Codebook design · Vector quantization · DEC network

Introduction

Background

Brain tumors are among the deadliest cancers, with an estimated nearly 300,000 new cases of brain and central nervous system cancers worldwide in 2020, ranking 19th in cancer incidence, according to the Global Cancer Observatory (https://gco.iarc.fr/today/online-analysis-table). Magnetic resonance imaging (MRI), a typical non-invasive imaging technique that produces high-quality brain images without injury and skull artifacts while providing more comprehensive information for brain tumors, is considered the main technical tool of diagnosing and treating brain tumors. Segmentation is a key step in biomedical image processing, aiming to provide a reliable basis for pathology research and assist doctors in making accurate clinical diagnoses.

The widely used segmentation methods may discard spatial information, and thus lose low-level semantics (Havaei et al. 2017; Wang et al. 2013). The importance of 3D volume data segmentation of MRI images has been

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widely recognized in the field of artificial intelligence-aided diagnosis (Dolz et al. 2020). Spatial information between slices has substantial meaning (the views of transverse, sagittal and coronal) for radiologists when reading MRI images. Recently, CNN-based systems, such as LeNet (LeCun et al. 1998), AlexNet (Krizhevsky et al. 2017), VggNet (Simonyan and Zisserman 2014), GoogleNet (Szegedy et al. 2015), ResNet (He et al. 2016), and DenseNet (Huang et al. 2017), have achieved great success and represented a series of breakthroughs in image segmentation. Application areas include brain MRI image segmentation (Dolz et al. 2020), lung segmentation (Altan and Karasu 2020), cell segmentation in electron microscope recordings (Al-Kofahi et al. 2009), dermoscopic image segmentation (Lin et al. 2017), COVID-19 detection and classification (Bhosale and Patnaik 2022, 2023, 2022), EEG classification (Liu et al. 2022), and other areas (Yu and Zhao 2023; Sezer and Altan 2021, 2021).

Prior works and problems

So far, researchers have proposed many image segmentation methods, most of which are only based on the gray value information of 2D data (Krinidis and Chatzis 2010; Kannan et al. 2012; Wang et al. 2013; Xu et al. 2013). Therefore, 3D segmentation for brain MRI images has received extensive attention (Kamnitsas et al. 2017; Fidon et al. 2017; Daimary et al. 2020; Zhou et al. 2020; Havaei et al. 2017; Lu et al. 2019; Derikvand and Khotanlou 2020; Samikannu et al. 2020; Mheich et al. 2015; Wang et al. 2019; Ortiz et al. 2013; Prastawa et al. 2005; Weisenfeld et al. 2006; Anbeek et al. 2008; Chen et al. 2018; Kamnitsas et al. 2017), such as brain tumors (Kamnitsas et al. 2017; Fidon et al. 2017; Daimary et al. 2020; Zhou et al. 2020; Havaei et al. 2017; Lu et al. 2019; Derikvand and Khotanlou 2020; Samikannu et al. 2020), brain tissues of adults (Mheich et al. 2015; Wang et al. 2019; Ortiz et al. 2013) and infants (Prastawa et al. 2005; Weisenfeld et al. 2006; Anbeek et al. 2008).

Recently, our community has witnessed the widespread adoption of deep learning-based models as effective alternatives to traditional segmentation approaches. Most architectures are supervised models trained end-to-end to learn the hierarchies of image features representing different levels of abstraction. Therefore, in terms of tumor detection, candidate generation and false positive reduction, deep learning-based methods significantly outperform traditional machine learning-based methods (Krinidis and Chatzis 2010; Kannan et al. 2012; Wang et al. 2013; Dhaniya and Umamaheswari 2022). This performance was validated in 2015 when DeepMedic, a software for brain lesion segmentation based on 3D CNN coupled with 3D fully connected CRF, won the ISLES 2015 challenge (Kamnitsas et al. 2015). Other deep learning-based brain tumor segmentation methods were also proposed in 2013, 2015 and 2017 challenges. In addition to 3D CNN (Wang et al. 2019; Zhang et al. 2019), many other deep learning models were adopted, including FCNN (Xu et al. 2013; Mheich et al. 2015), FCNN with CRF (Shehab et al. 2015), 3D U-Net (Havaei et al. 2016; Kamnitsas et al. 2017) and Autoencoders (Fidon et al. 2017; Prastawa et al. 2005).

As more studies have been devoted to 3D deep learning-based segmentation, derivatives and improvements of classic types of CNN models have been continuously reported, as summarized in Table 1. U-Net is one of the most popular models in medical image segmentation with a deeply supervised architecture. In Li and Ren (2021), attention mechanisms are introduced to make U-Net network focus on the key feature regions during network training, and experiments confirm that the new model has better outcomes than the fully convolutional network (FCN) and U-Net. AGResU-Net integrates residual modules and attention gates with a primeval and single U-Net architecture, pays more attention to the information of small-scale brain tumors, and outperforms the baselines of U-Net and ResU-Net (Zhang et al. 2020). Meanwhile, more and more researchers are beginning to realize the importance of computational efficiency in deep learning network systems. A CNN-based INet model and its developed version, named Dense-INet were proposed to perform segmentation by gradually increasing kernel sizes of convolutional layers instead of low-level semantics (Weng and Zhu 2021). The results demonstrate that the INet and DenseINet require 16.9% and 37.6% fewer parameters than ResUNet and ResDenseUNet, respectively, and save a lot of calculation power.

To sum up, CNN-based methods achieved state-of-the-art performances compared to other methods, as shown in Table 1, on a broad range of medical image segmentation problems (Nie et al. 2016; Dolz et al. 2020; Li and Li 2011; Cibi and Rose 2022), including some multi-modal tasks (Wang et al. 2011; Srhoj-Egekher et al. 2012; Chen et al. 2018; Deoni et al. 2007). These advanced models have been applied in both theoretical research and clinical treatment assistance. However, after years of processing real medical images in hospital facilities, there are still significant obstacles to implementing theoretically perfect models to real clinical datasets. First, CNN-based models mostly work under supervision that invariantly requires large amounts of labelled data for training, which is typically rarely available in medical imaging communities due to patient confidentiality or other restrictions. Unsupervised deep learning networks would be a suitable solution for practical medical applications. Furthermore, as a validated 3D CNN-based model, the network architecture is quite deep and complex, leading to the problem of computational

Table 1 Overview of representative works on brain segmentation

Work	Method	Data	Target
Prastawa et al. (2005)	Multi-atlas	T1, T2	Infant brain
Weisenfeld et al. (2006)	Multi-atlas	T1, T2	Infant brain
Deoni et al. (2007)	k-means	T1, T2	Infant brain
Anbeek et al. (2008)	KNN	T2, IR	Infant brain
Ortiz et al. (2013)	Multi-atlas	T1, T2	Infant brain
Wang et al. (2011)	Multi-atlas, level set	T1, T2	Infant brain
Srhoj-Egekher et al. (2012)	Multi-atlas, KNN	T1, T2, FA	Infant brain
Wang et al. (2012)	Multi-atlas	T1, T2, DWI, Flair	Infant brain
Wang et al. (2014)	Multi-atlas, level set	T1, T2, FA	Infant brain
Kamnitsas et al. (2015)	3D FCNN	T1, T2, Flair	Brain
Zhang et al. (2015)	2D CNN	T1, T2, FA	Infant brain
Havaei et al. (2016)	2D FCNN	T1, T2, Flair	Tumor
Nie et al. (2016)	3D FCNN	T1, T2, FA	Infant brain
Chen et al. (2018)	3D FCNN	T1, T2, Flair	Brain
Dolz et al. (2020)	CNN	T1, T2	Infant brain
Fidon et al. (2017)	3D FCNN, CRF	T1, T2, Flair	Tumor
Kamnitsas et al. (2017)	3D FCNN	T1, T2, Flair, PD	Tumor
Valverde et al. (2017)	3D FCNN	T1, T2, Flair	Injury
Kamnitsas et al. (2017)	3D CNN, CRF	T1, T2, Flair	Tumor
Daimary et al. (2020)	Hybrid CNN	T1, T2, Flair	Tumor
Zhou et al. (2020)	Dense CNN	T1, T2, Flair	Tumor
Havaei et al. (2017)	Two-path CNN	T1, T2, Flair	Tumor
Lu et al. (2019)	Multi-path 3D CNN	T1, T2, Flair	Tumor
Derikvand and Khotanlou (2020)	Hybrid CNN	T1, T2, Flair	Tumor
Samikannu et al. (2020)	CNN	T2	Glioma
Li and Ren (2021)	3D U-Net	T1, T2	Tumor
Zhang et al. (2020)	ResU-Net	T1, T2	Tumor
Weng and Zhu (2021)	INet	T1, T2, Flair	Tumor

efficiency. Hence, in order to guarantee the validity and robustness of the model, the efficiency cost is always overwhelming. Therefore, a 3D brain data segmentation system with appropriate speedup strategy is indispensable.

Our proposal and contributions

In this paper, we propose a vector quantization (VQ)-based 3D segmentation method for brain MRI data, in which a novel unsupervised deep learning approach is employed to implement the codebook design for VQ, and a memory-reserved strategy is developed to realize the 3D deep network efficiently. For the first problem, we introduce the idea of segmentation based on VQ, and propose an unsupervised AI model of 3D deep embedding clustering (DEC) network to enforce the VQ process (Xie et al. 2016; Huang et al. 2014). In particular, for real brain datasets, the 3D deep learning network is trained in an unsupervised manner using the designed architecture to ensure that the model achieves accurate segmentation without proper annotation preprocessing. For the second problem, based on the characteristics of medical data, a memory reserving-and-fading strategy is developed to implement the fast codebook design for efficient training of 3D-DEC networks. The strategy explores the dependencies between slices as well as the consistency of shape and size, aiming to efficiently acquire essential information to speedup training time. Specifically, a typical slice of MRI images is determined according to the region of interest (ROI, such as tumor location), and network connections of adjacent slices are enhanced and reserved, while network connections of remote slices are weakened and faded. Therefore, this strategy greatly reduces the computational complexity burden while largely preserving the most relevant information. Experimental results on three brain image databases (IBSR, BrainWeb, and real MRI data) demonstrate that the proposed unsupervised deep learning model and its efficient training strategy are beneficial for the brain tumor segmentation task. In general, the main contributions of this paper are summarized as follows:

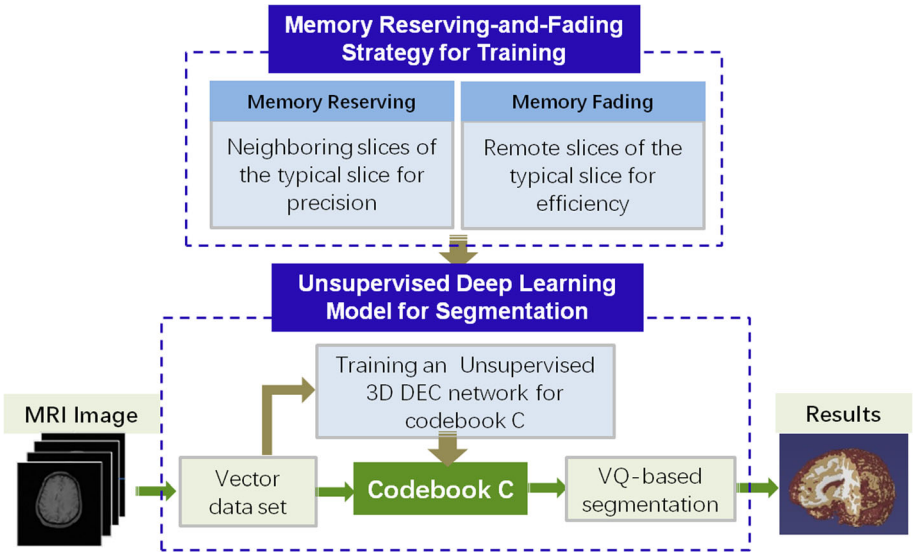


Fig. 1 Block diagram of the VQ-based 3D segmentation method

(1) To address the manual annotation problem in real MRI brain data processing, we designed 3D-DEC network trained in an unsupervised manner based on VQ technique, which is able to achieve accurate segmentation without proper annotation preprocessing.
(2) By introducing the memory reserving-and-fading strategy in codebook design, the proposed 3D-DEC network can greatly reduce the computation complexity while retaining the most relevant information, which is particularly feasible for complex deep learning system with large amount of dataset.
(3) Segmentation experiments were conducted on two baseline databases (IBSR and BrainWeb) and a real brain MRI database from the hospital. Results confirmed that the proposed 3D-DEC network can achieve competitive segmentation accuracy with high speedup compared to state-of-the-art 3D CNN models. Therefore, potentially complicated clinical diagnosis or treatment becomes possible, by introducing the Computer-Aided Diagnosis system based on the proposed model in the future.

The rest of this paper is organized as follows: Sect. 2.1 describes the overall scheme of the proposed VQ-based segmentation method and the vector set construction. Section 2.2 presents the VQ-based segmentation algorithm. Section 2.3 describes the details of the efficient memory reserving-and-fading strategy for 3D deep network training. Section 2.4 introduces the codebook learning program implemented by the unsupervised deep neural network (DEC-net). Section 3 shows experimental results of our work on simulated brain MRI data and real tumor MRI data. Discussion and conclusion are given in Sects. 4 and 5.

Methods

In this study, we propose a VQ-based 3D segmentation method for brain MRI images. The key step of this method is codebook design. We introduce an unsupervised 3D-DEC model to process voxel-wise volume data, and combine the memory reserving-and-fading strategy with codebook training enforced by the DEC network. The overall scheme is illustrated in Fig. 1.

Dataset construction

As shown in Fig. 1, the vector dataset needs to be trained with a 3D DEC neural network to obtain the codebook. First, the vector set X is constructed by subcubes of the neighboring voxel regions in a certain form. Then, the vectors of X are quantized into K_C (the number of codewords in C) classes according to the codebook C obtained from the trained 3D-DEC network. Figure 2a shows a set of MRI volume images, and their axial, coronal and sagittal views. Figure 2b shows the corresponding 2D matrices, 3D data set, and the formation of a subcube.

The proposed method introduces the spatial information of the volume image along the z-axis direction in addition to the directions of x- and y-axes. Therefore, during the vector construction, the 3D MRI image is divided into overlapping subcubes of $n \times n \times n$ voxels. Each subcube is converted to a vector of $n \times n \times n$ elements, and then a vector set X is constructed for each volume data.

Suppose that $V = \{f(i,j,l)\}$ represents the 3D data of an $M \times N \times H$ volume image $[f(i,j,l)]_{M\times N\times H}$, and $[f(i,j,l)]_{n\times n\times n} = (i = i_k, i_k+1, \ldots, i_k+n-1; j = j_k, j_k+1, \ldots, j_k+n-1; l = l_k, l_k+1, \ldots, l_k+n-1)$ is the kth subcube of the volume image. The subcube can be

expressed as vector $X(k)$, and $X(k) = [f(i_k, j_k, l_k), \ldots, f(i_k, j_k + n - 1, l_k), \ldots, f(i_k + n - 1, j_k + n - 1, l_k), \ldots, f(i_k + n - 1, j_k + n - 1, l_k + n - 1)]^T$. Therefore, a 3D image $[f(i,j,l)]_{M\times N\times H}$ can be represented by a vector set of $X = \{X(k); k = 1, 2, \ldots, K_X\}$, where $K_X = M \times N \times H$ is the total number of voxels.

VQ-based segmentation algorithm

In this study, we perform volume segmentation based on the VQ technique, and obtain the codebook C using an unsupervised deep learning model. Finally, the vector set is segmented by the VQ-based algorithm, as shown in Fig. 3.

The VQ technique is basically a mapping process as: $X \in R^m \rightarrow C \in R^m$. $C = \{C(k); k = 1, 2, \ldots, K_C\}$ represents the codebook, $C(k)$ represents the code word, and K_C is the number of clusters (Ge et al. 2013; Dimitrovski et al. 2014). As described earlier in Sect. 2.1, the 3D volume data $[f(i,j,l)]_{M\times N\times H}$ has been reconstructed into a vector set form $X = \{X(k); k = 1, 2, \ldots, K_X\}$, $K_X = M \times N \times H$. Given the codebook $C = \{C(k); k = 1, 2, \ldots, K_C\}$, the segmentation algorithm for vector set $X(k)$ is summarized as follows:

(i) The Euclidean distance between $X(k)$ and each code word $C(\tilde{k})$ of the codebook C is obtained according to:

$$d_{\tilde{k}}(k) = \|X(k) - C(\tilde{k})\|, \quad (1)$$

where $\tilde{k} = 1, 2, \ldots, K_C$.

(ii) The Euclidean distance is optimized and the minimum distance is obtained as follows:

$$\underset{j}{\arg\min}\{d_j(k)\}. \quad (2)$$

(iii) Suppose $C(q)$ represents the code word that has the minimum distance to $X(k)$, the vector $X(k)$ is classified to the clusters represented by $C(q)$. Then, the subcube $[f(i,j,l)]_{n\times n\times n}$ is segmented.

(iv) The above steps are repeated until all non-edge vectors are segmented to K_C clusters, where each cluster corresponds to a code word $C(\tilde{k})$ of C (De et al. 2016).

Memory reserving-and-fading strategy

Depending on the correlation between the current processing slice and the "typical slice", different normalization methods are employed to implement the memory reserving-and-fading strategy for deep network training. Generally, the "typical slice" can be chosen by the center of the volume. For volume images of tumor patients in clinical applications, the selection of "typical slice" can be designed according to the location of the tumor. Remote slices and neighboring slices are divided according to application context. For example, for brain images with tumors, the tumor portion should be divided into neighboring slices.

The batch normalization (BN) has been adopted to fix training problem of deep networks, and it can accelerate

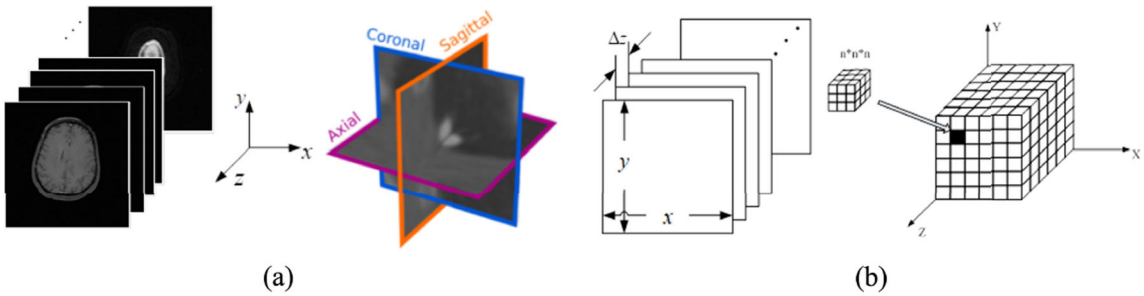


Fig. 2 3D volume data constitution and the formation of a subcube

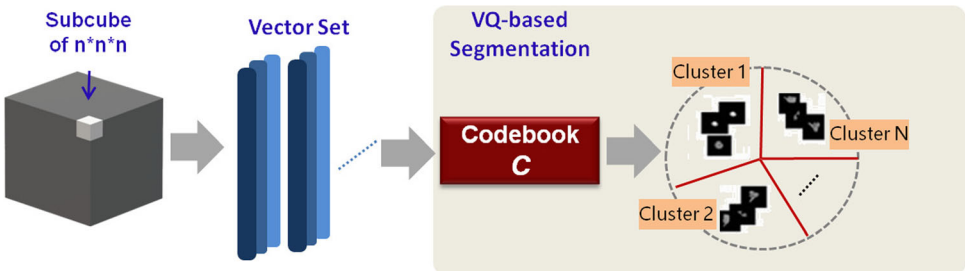


Fig. 3 Schematic diagram of VQ-based volume segmentation

the training process by adjusting the learning rate. Since BN mitigates the insensitivity to the initial weight of networks, the obtained codebook C of "typical slice" can be used as the initialization for the next slice, and the subsequent slices can be initialized in the same way. The neighbouring slices can be trained with major training speedup without side effects on accuracy. Hence, the proposed deep network model with BN yields a substantial speedup with only a small fraction of training steps by setting a higher learning rate.

In this part, we employ BN for reparameterization in deep learning networks to achieve memory-reserving for neighboring slices. BN can be used to enhance the training performance of each epoch through data normalization, and the reparametrized data $\hat{x_i}$ can be presented as

$$\hat{x_i} = \gamma \frac{x_i - \mu}{\sigma} + \beta, \tag{3}$$

where x_i denotes the original data.γ and β are adjusting parameters (Wang et al. 2012). μ and σ represent mean and standard deviation.

We use weight normalization (WN) to achieve memory-fading for remote slices, where WN is a reparameterization used to tune training convergence by adjusting the gradient descent. In this way, the minimum noise is introduced, which in turn preserves the maximum algorithm accuracy. The weight vector ω is reparameterized by a vector v and a scalar g, which are updated by the batch gradient descent (BGD) algorithm (Zhang et al. 2015; Nie et al. 2016),

$$w = \frac{g}{\|v\|} v. \tag{4}$$

In summary, we developed an efficient method for codebook computation based on the chosen "typical slice" of MRI volume images. Reparameterizations of BN and WN were adopted in deep network training for the most relevant neighboring slices and remote slices, respectively. The final codebook C for brain MRI volume images was calculated After rapid convergence. The overall framework of the memory reserving-and-fading strategy is shown in Fig. 4.

Codebook learning by 3D-DEC network

In this part, we build an unsupervised 3D-DEC model for the codebook learning task (Xie et al. 2016) and make the learned representations suitable for clustering, as shown in Fig. 5. The 3D-DEC model is upgraded by the foundational deep autoencoder (DAE) architecture by introducing two constraints on learned representations. A locality-persevering constraint is imposed to embed the original data into its underlying manifold space, and a group sparsity constraint is imposed to learn block diagonal representations with non-zero groups corresponding to its clusters.

For the deep network, the input vector can be expressed as $x = [x_1, x_2, \ldots, x_{n^3}]^T$, W is the weight. The encoder defines a transformation $f(\cdot)$ that transforms the input x to $f(x)$

$$f(x) = W_3^T \phi\big(W_2^T \phi\big(W_1^T x\big)\big), \tag{5}$$

where $\phi(\cdot) = 1/(1 + e^{-x})$ is the activation function. For simplicity, we drop the bias term for each layer in the formulation.

Unrolling the four-layer encoder network, we obtain a deep autoencoder with a symmetric decoder. The decoder also defines a transformation $\hat{f}(\cdot)$, which uses the transformed representation $f(x)$ to reconstruct the original input x. The reconstructed input is

$$\hat{f}(x) = W_3^T \phi\big(W_2^T \phi\big(W_1^T f(x)\big)\big). \tag{6}$$

The regular objective function of the proposed network can be written as

$$E_r = \sum_{i=1}^{K_X} \|x_i - \hat{f}(x_i)\|^2, \tag{7}$$

where $\|\cdot\|$ denotes the Euclidean norm.

The proposed 3D DEC model utilizes a deep network to find the reduced features from the 3D volume dataset. Next, we aim to learn clustering-oriented representations by imposing locality-preserving constraints and group sparsity constraints.

Locality-preserving Constraint

In order to make the learned representations suitable for clustering, constraints should be embedded into their intrinsic manifold to preserve their local property. We introduce a locality-preserving constraint as follows,

$$E_g = \sum_{i,j \in k(i)} S_{ij} \|f(x_i) - f(x_j)\|^2, \tag{8}$$

where $k(i)$ is the set containing the indexes of k nearest neighbours of data x_i. $S_{ij} = e^{-\|x_i - x_j\|^2/t}$ is the similarity measurement, and t is a tuning parameter.

Group sparsity constraint

Inspired by spectral clustering that exploits block diagonal similarity matrix for representation learning, we aim to diagonalize the learned representations by selecting one or more relevant groups of hidden units that represents inherent properties of the data to further facilitate clustering. Next, we employ group lasso which leads to sparsity within hidden codes on group level to find the block

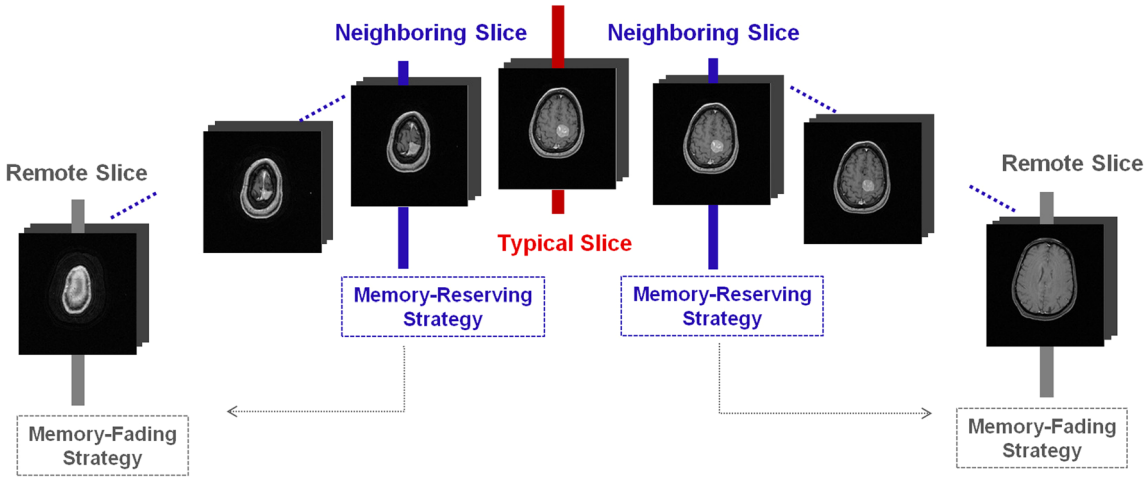


Fig. 4 Schematic diagram of memory reserving-and-fading strategy

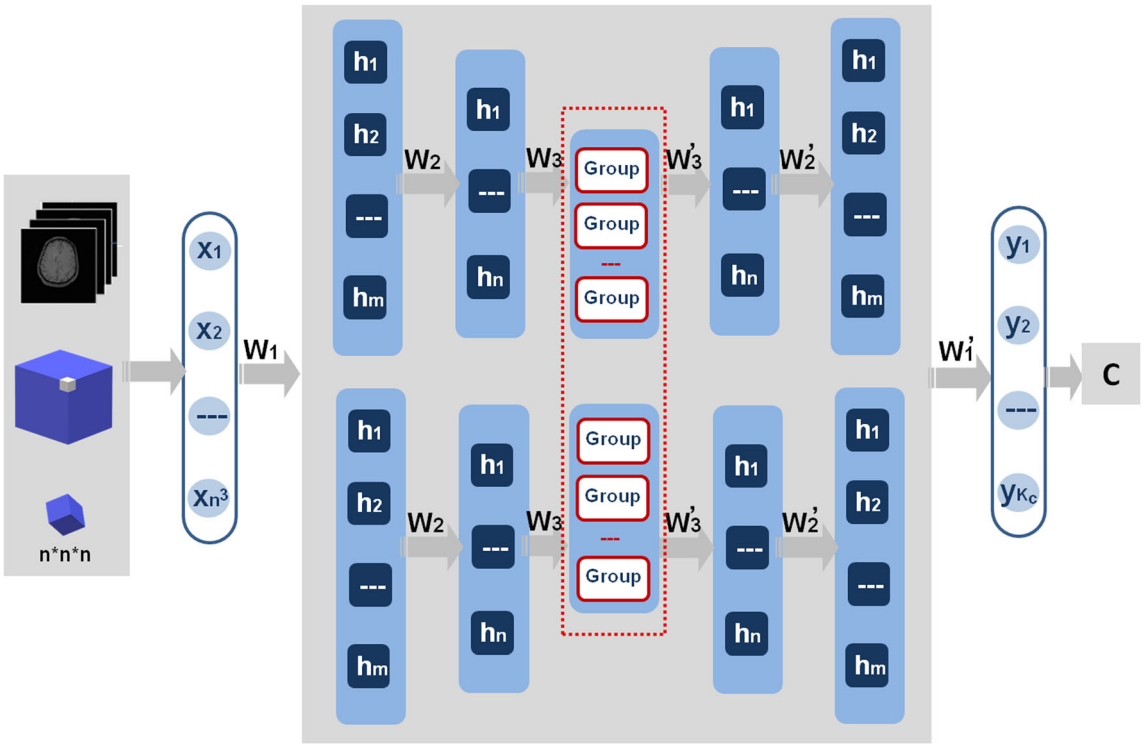


Fig. 5 Network architecture of the proposed DEC network

diagonal features. The hidden units are divided into K_C groups, and K_C is the number of clusters. Then, the group sparsity constraint can be imposed as follows,

$$E_s = \sum_{i=1}^{K_X} \sum_{j=1}^{K_C} \lambda_j \| f^j(X_i) \|, \tag{9}$$

where $[\lambda_j]_{j=1}^{K_C}$ is the weight of the group sparsity constraint, and K_X is total voxel number of the MRI image. Generally, larger weights are given to larger groups and λ_j is defined as $\lambda_j = \lambda\sqrt{n_j}$, where n_j is the group size and λ is a predefined constant.

By imposing the group sparsity constraint, only a small number of groups can be activated in the learned representation of each data point, and the activated groups correspond to specific clusters. As a result, all representations can be block-diagonalized.

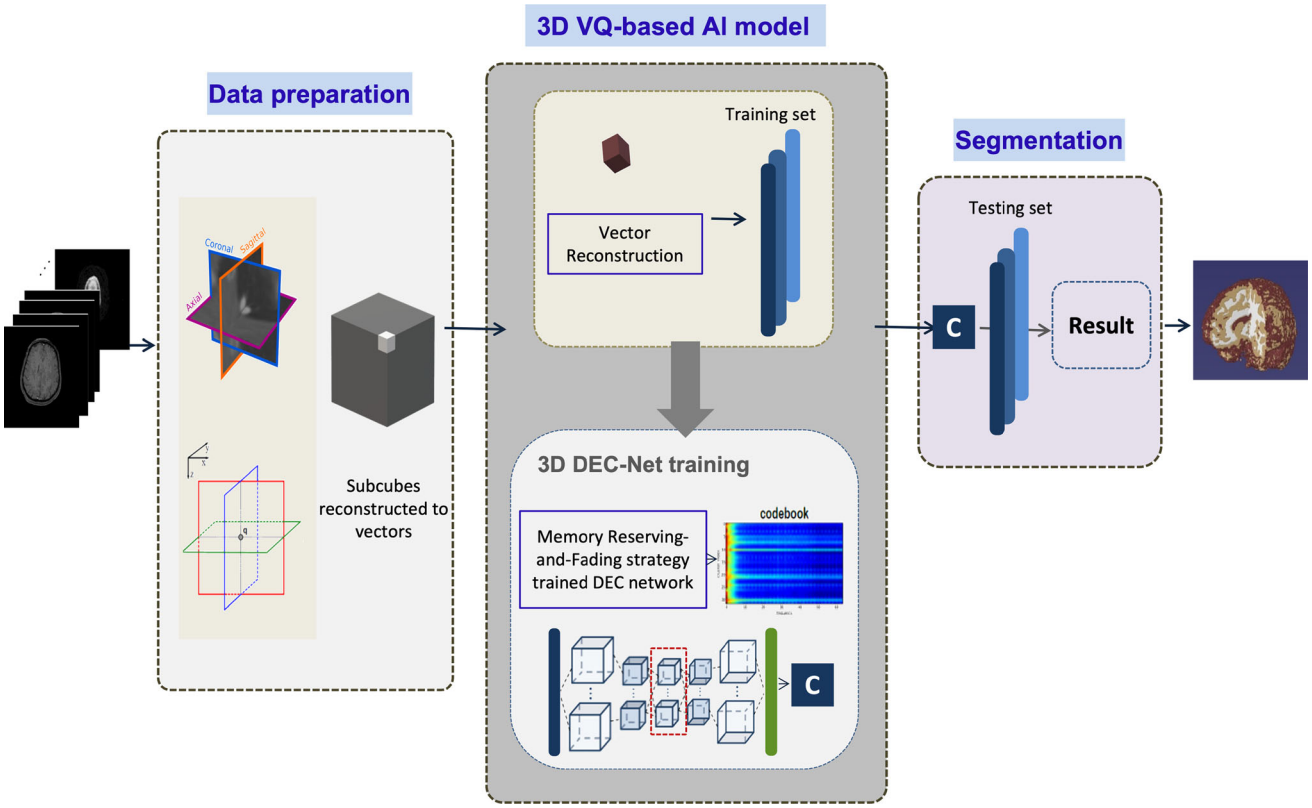


Fig. 6 Overall schematic architecture of the proposed algorithm

Overall objective function

After introducing the above constraints, the overall objective function of the proposed DEC network is expressed as follows

$$E = E_r + \alpha E_g + \beta E_s, \tag{10}$$

where α and β are adjusting parameters (Valverde et al. 2017).

As shown in Fig. 5, the weights $W_j = [W_{j1}, W_{j2}, W_{j3}, W'_{j3}, W'_{j2}, W'_{j1}]^T$, $j = 1, 2, \ldots, K_C$ are trained using the descending gradient method. In addition, the objective function is optimized using the backpropagation algorithm (Valverde et al. 2017). After the training process converges, the computed weight $W'_1 = \{W'_1(j), j = 1, 2, \ldots, K_C\}$ is equal to the codebook C as discussed in Sect. 2.2.

Figure 6 illustrates the overall schematic architecture of the proposed model.

Experiments and results

In this study, we first used simulated brain MRI images from two public baseline medical databases, including IBSR[1] (18 samples with volume size of $256 * 256 * 128$) and BrainWeb[2] (18 samples with volume size of $181 * 181 * 217$), to verify the accuracy and efficiency of segmentations. Then, we used a new database created by real brain MRI images (56 samples with volume size of $256 * 256 * 28$) with tumors from our institute for clinical application.

All experiment results were the average of 100 independent runs and each run underwent 5-fold cross validation.

Experiments using simulated normal brain MRI volume images

In this section, we use the simulated brain MRI volume images from IBSR and BrainWeb to verify the segmentation precision and efficiency of the proposed method.

[1] https://www.nitrc.org/projects/ibsr/.

[2] https://brainweb.bic.mni.mcgill.ca/.

Table 2 Performance comparison of Tanimoto parameter on IBSR database

Method	White matter	Gray matter	Cerebrospinal fluid
EGS-SOM (Ortiz et al. 2013)	0.70 ± 0.04	0.70 ± 0.04	0.22 ± 0.08
HFS-SOM (Ortiz et al. 2013)	0.60 ± 0.10	0.60 ± 0.15	0.10 ± 0.05
3D CNN (Kamnitsas et al. 2017)	0.73 ± 0.04	0.74 ± 0.05	0.24 ± 0.05
3D U-Net (Ronneberger et al. 2015)	0.74 ± 0.04	0.74 ± 0.05	0.22 ± 0.04
Ours	0.74 ± 0.04	0.74 ± 0.04	0.24 ± 0.05

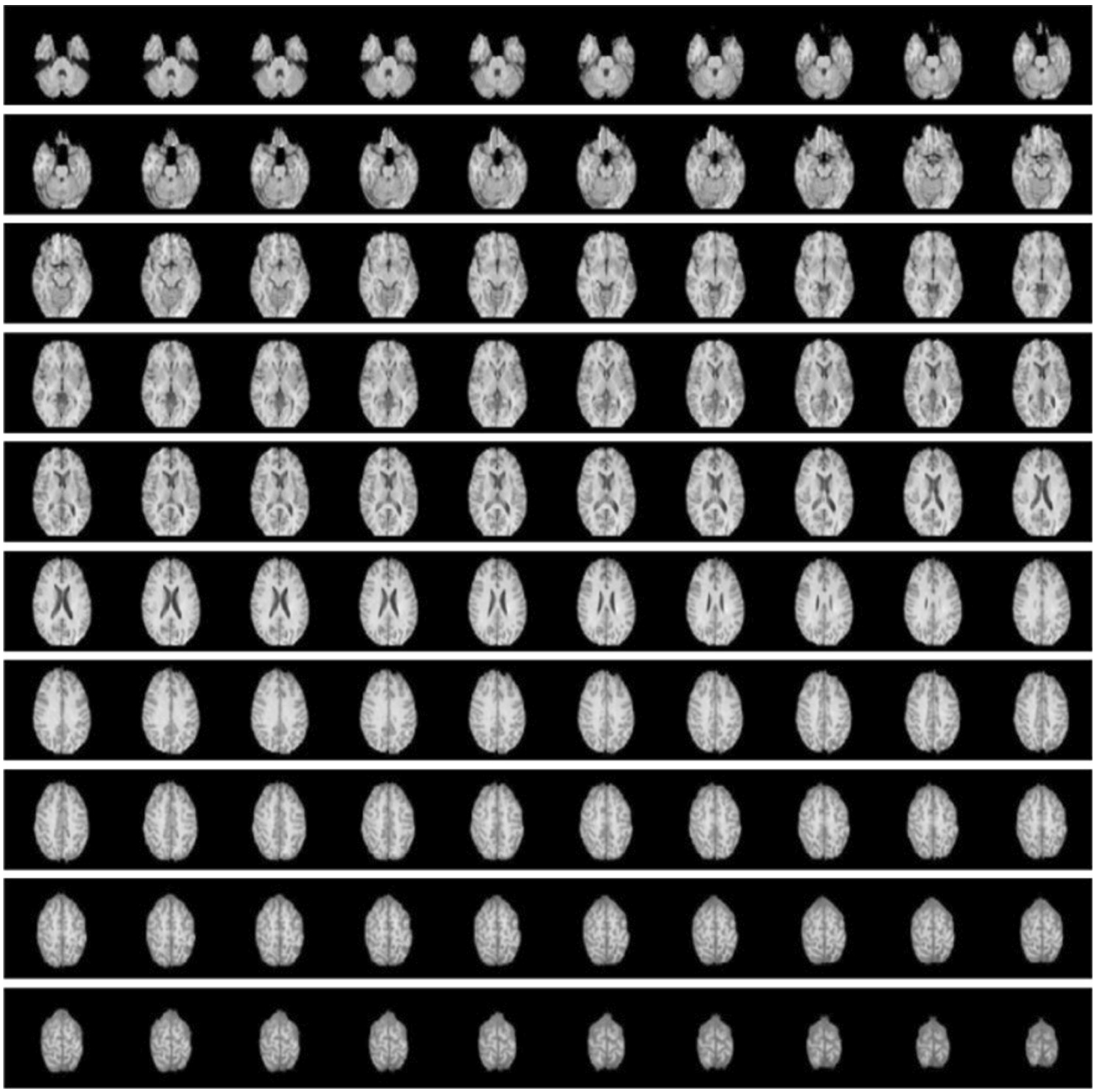


Fig. 7 Image samples of Volume#12 from IBSR

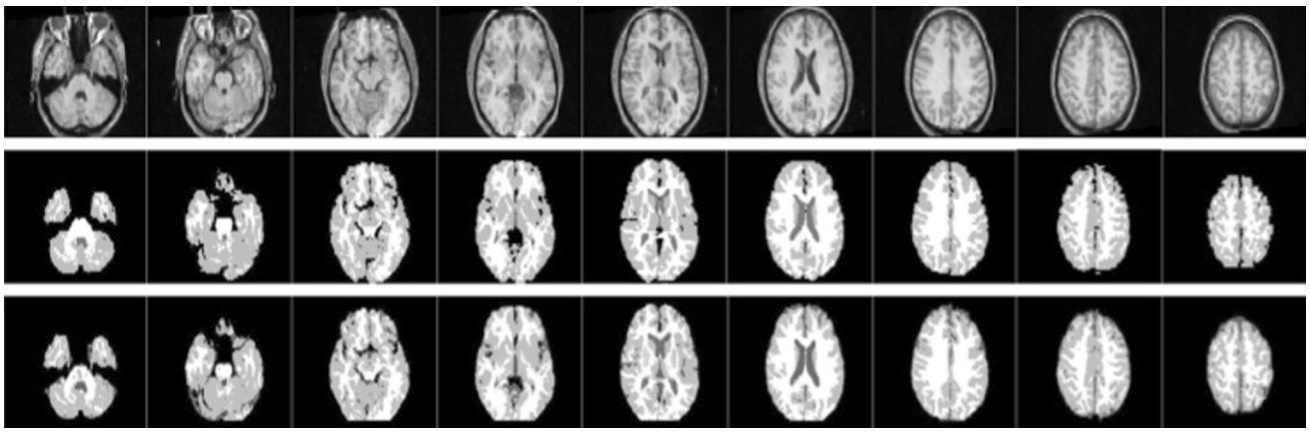


Fig. 8 Segmentation results of sample Volume#12 from the IBSR using the proposed method. From left to right shows slice 100, 110, 120, 130, 140, 150, 160, 170 and 180. Original images are shown in the first row, ground truths are shown in the second row, and segmented results are shown in the third row

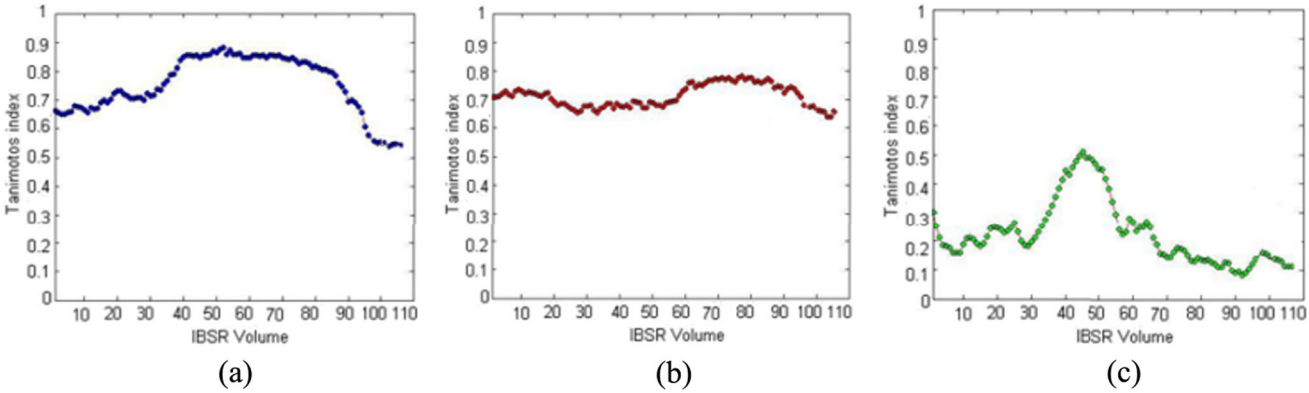


Fig. 9 Tanimoto index of segmentation result of our method for Volume#12. **a** White matter; **b** Gray matter; **c** Cerebrospinal fluid

Evaluation criteria

First, we introduce the tanimoto parameter to verify the precision of the proposed segmentation method, which is defined as

$$T(S_1, S_2) = \frac{|S_1 \bigcap S_2|}{|S_1 \bigcup S_2|}, \tag{11}$$

where S_1 and S_2 represent the segmented result and the corresponding golden standard, respectively.

Second, we adopte four similarity parameters of TP (true positive), TN (true negative), FN (false negative), and FP (false positive) to evaluate the performance of the proposed method as follows:

$$\text{TP} = |TP|/|T|, TP = S \cap T, \tag{12}$$

$$\text{FN} = |FN|/|T|, FN = T - (S \cap T), \tag{13}$$

$$\text{FP} = |FP|/|I - T|, FP = S - (S \cap T), \tag{14}$$

$$\text{TN} = |TN|/|I - T|, TN = I - (S \cup T). \tag{15}$$

Third, we measure the spatial overlap, boundary distance and volume using dice similarity coefficient (DSC), modified Hausdorff distance(MHD) and average symmetric surface distance (ASD), respectively.

IBSR database

In this part, we assess the performance of our work with four different baselines including EGS-SOM (Ortiz et al. 2013), HFS-SOM (Ortiz et al. 2013),3D CNN (Kamnitsas et al. 2017) and 3D U-Net (Ronneberger et al. 2015). Regarding the segmented white matter, gray matter and cerebrospinal fluid, Table 2 gives the Tanimoto parameter comparison.

To better illustrate the segmentation performance of our work, we choose the Volume#12 as an example, parts of which are shown in Fig. 7. The entire volume dataset contains 128 slices of brain images. After the basic processing, the size of each subcube is set to $3 \times 3 \times 3$ (n = 3). Figure 8 shows segmentation results of some slices of Volume#12, and Fig. 9 shows the Tanimoto index of

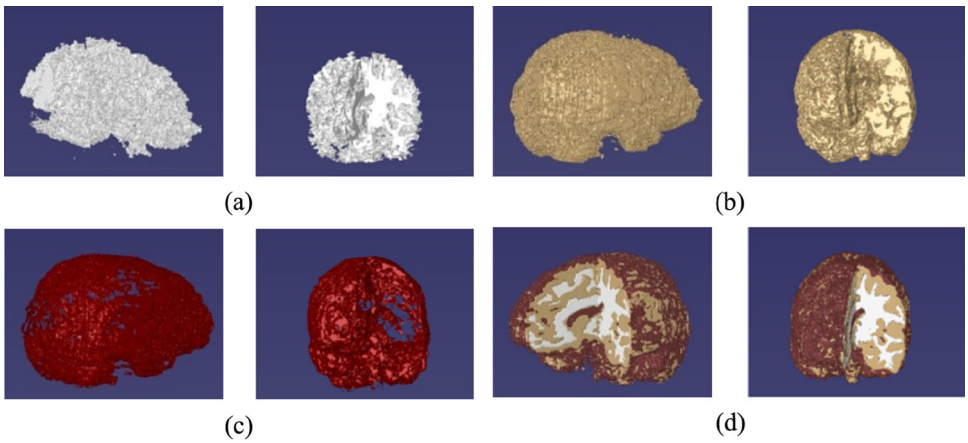


Fig. 10 Segmentation illustration of Volume#12. **a** White matter; **b** Gray matter; **c** Cerebrospinal fluid; **d** All segmented parts shown together

segmentation result. The 3D visualizations of an example segmentation result are shown in Fig. 10. Figure 10a–d illustrate the segmented white matter, the segmented gray matter, the segmented cerebrospinal fluid, and the combination of all segmented parts.

BrainWeb database

For broad comparison, we first evaluate the performance (TP, FN, FP and TN values of white matter and gray matter) of our work with five baselines including three clustering algorithms of FLICM (Krinidis and Chatzis 2010), SFCM (Kannan et al. 2012) and ASIFCM (Wang et al. 2013), and two deep networks of 3D CNN (Kamnitsas et al. 2017) and 3D U-Net (Ronneberger et al. 2015), as shown in Table 3. Then, we compare the results on DSC, MHD and ASD values with deep training related works including DeepMedic (Roth et al. 2018), 3D ResNet (Sun et al. 2019), 3D CNN (Kamnitsas et al. 2017) and 3D U-Net (Ronneberger et al. 2015), as shown in Table 4.

We take the volume image samples of Subject#5 as an example, as shown in Fig. 11. The whole volume data set contains 217 slices. Similarly, the size of each subcube is set to $3 \times 3 \times 3$. Figure 12 shows some of segmentation results of Subject#5 using the proposed method. The 3D visual results of the proposed method for sample Subject#5 are shown in Fig. 13. Figure 13a–c illustrate the segmented white matter, the segmented gray matter, and the combination of all segmented parts. To verify the merits of the memory reserving-and-fading strategy, experimental tests are conducted to confirm the computational efficiency. The computation time to train a deep neural network to obtain a codebook using our work and another deep network (3D CNN, Kamnitsas et al. 2017) is shown in Table 5.

Experiments using real brain MRI volume images

In practice, the segmentation of real medical images, especially those containing tumors, is more meaningful In clinical applications than that of simulated images. Therefore, in this section, we conduct the proposed method on tumor MRI volume images acquired in the clinic.

Evaluation criteria

To evaluate the performance of the propose model, we adopt three standard performance metrics: Dice score, sensitivity and specificity. Dice score measures the similarity between samples, sensitivity measures the proportion of true positives that are correctly identified and specificity

Table 3 Performance comparison of TP, TN, FN, and FP on BrainWeb database

Method	White matter				Gray matter			
	TP	FN	FP	TN	TP	FN	FP	TN
FLICM (Krinidis and Chatzis 2010)	93.7	6.3	5.6	94.4	95.4	4.6	3.9	96.1
SFCM (Kannan et al. 2012)	93.0	7.0	4.5	95.5	95.5	4.5	3.1	96.9
ASIFCM (Wang et al. 2013)	94.2	5.8	5.1	94.9	93.3	6.7	4.2	95.8
3D CNN (Kamnitsas et al. 2017)	97.4	2.6	2.4	97.6	98.0	2.0	2.2	97.8
3D U-Net (Ronneberger et al. 2015)	97.4	2.6	2.3	97.7	97.8	2.2	2.2	97.8
Ours	**97.5**	**2.5**	**2.3**	**97.7**	**98.0**	**2.0**	**2.1**	**97.9**

*TP: True positive, TN: true negative, FN: false negative, and FP: false positive

Bold is indicate the optimal performance in each comparison experiment

Table 4 Performance comparison of DSC, MHD and ASD on BrainWeb database

Method	White matter			Gray matter		
	DSC	MHD	ASD	DSC	MHD	ASD
DeepMedic (Roth et al. 2018)	0.86	7.01	0.411	0.85	6.45	0.342
3D ResNet (Sun et al. 2019)	0.86	6.78	0.378	0.86	6.41	0.338
3D CNN (Kamnitsas et al. 2017)	0.87	6.81	0.381	0.88	5.98	0.330
3D U-Net (Ronneberger et al. 2015)	0.88	6.67	0.374	0.87	5.88	0.367
Ours	**0.92**	**6.44**	**0.361**	**0.91**	**5.74**	**0.311**

*DSC: Dice similarity coefficient, MHD: modified Hausdorff distance, ASD: average symmetric surface distance

Bold is indicate the optimal performance in each comparison experiment

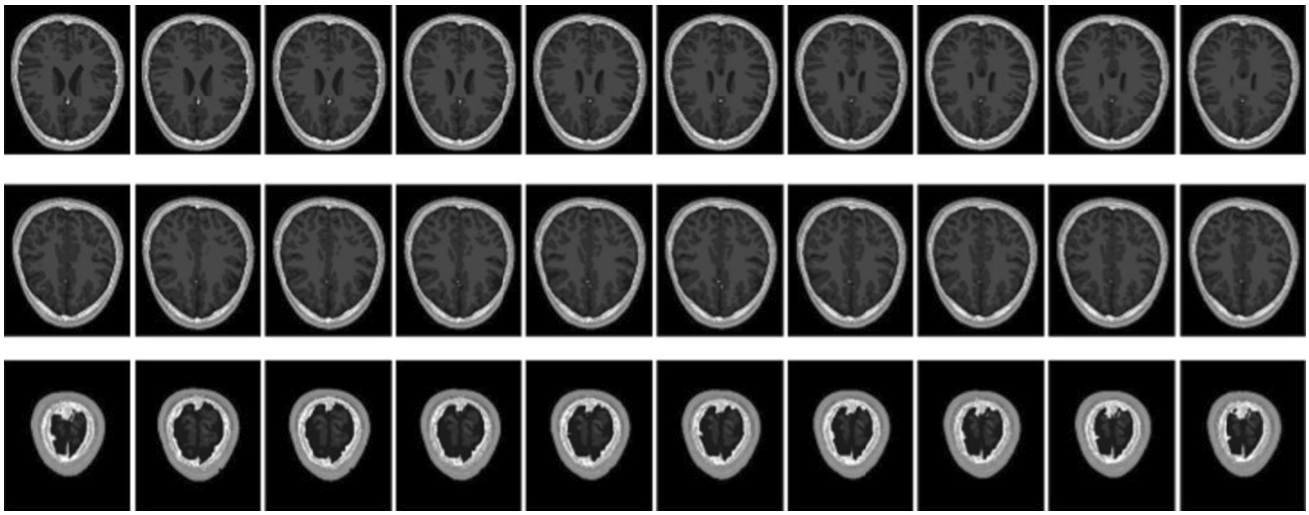


Fig. 11 Illustrations of 30 slices from Subject#5. The first row shows slices 192-201, the second row shows slices 216-225, and the third row shows slices 296-305

corresponds the proportion of true negatives that are correctly identified, which are calculated as follows:

$$\text{Dics} = \frac{|P_1 \cap T_1|}{(|P_1| + |T_1|)/2}, \tag{16}$$

$$\text{Sensitivity} = \frac{|P_1 \cap T_1|}{|T_1|}, \tag{17}$$

$$\text{Specificity} = \frac{|P_0 \cap T_0|}{|T_0|}, \tag{18}$$

where P denotes the segmented region and T denotes the ground truth. We use the annotations of three senior radiologists as the ground truth. Here P_1 and P_0 are the predicted positive and negative tumor regions respectively. Analogously, T_1 and T_0 are defined.

Database of real brain images with tumors

In this part, we compare the performance of our work with several state-of-the-art methods regarding the extracted tumor regions. Table 6 presents the quantitative results of the models, and performance comparisons in terms of Dice score, Sensitivity and Specificity. Figure 14 shows an example of brain image segmentation and tumor extraction. Figure 14a–d give illustrations of the original image, ground truth, extracted tumor and segmentation result of other parts.

To illustrate the experiment results, the volume image Sample#1 is selected as shown in Fig. 15. The whole volume data set contains 28 slices, and the size of each subcube is $3 \times 3 \times 3$. Figure 16 shows partial segmentation result of Sample#1 using the proposed method. The 3D visualizations of some results for are shown in Fig. 17. Figure 17a–e shows the segmented tumor, white matter, gray matter, skull and the combination result of all the segmented parts.

For further clinical application, our method can be used in the follow-up of tumor patient, and then tumor information can be studied periodically (e.g., every 6 months), thus the crucial details for surgical and postoperative treatments can be obtained by analyzing the numerical data

of tumors. Therefore, two additional samples from the same patient at different time periods are analyzed to complete this experiment. The former sample (named Sample#0) is the MRI volume images taken 6 months before Sample#1, and the latter Sample (named Sample#2) is taken 18 months after Sample#1. The volume images and their tumor segmentation results are shown in Figs. 18 and 19. Figure 20 shows the tumor variation status in three periods.

For the ablation study, the Dice, Sensitivity, Specificity, and running time of our proposed model (with and without speedup strategy) and 3D CNN model (without speedup strategy) were calculated on the real brain tumor database, as shown in Table 7. Extra comparative experiments are conducted on real database to validate the training efficiency of the memory reserving-and-fading strategy. By adding our strategy to traditional deep network training at different depths, we can see that our method proves to have a considerable advantage in efficiency over the no-strategy method, as shown in Fig. 21.

Furthermore, in experiments, we manually select the architecture for each deep embedding network, including the number of layers and the number of units in each layer, according to datasets and cluster numbers. By unrolling the four-layer encoder network, a deep autoencoder with a symmetric decoder is obtained, and Fig. 5 shows the architecture of the proposed 3D-DEC network. Table 8 provides the parameter settings and computational complexity for the proposed architecture. We fine-tune the

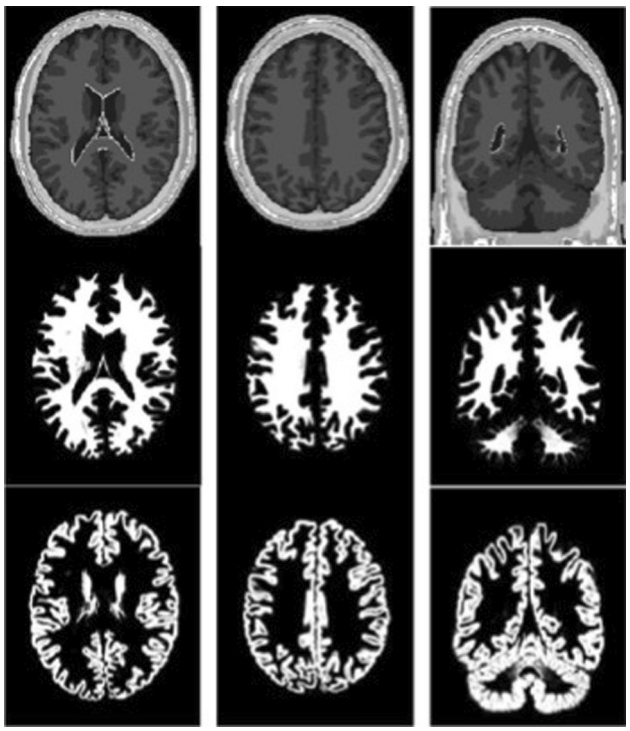


Fig. 12 Segmentation results of volume Subject#5. Original images, white matters and gray matters extracted by proposed method are shown in rows from up to bottom

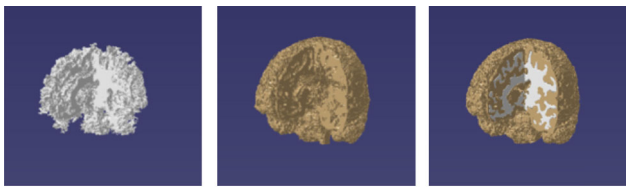


Fig. 13 Segmentation illustration of volume Subject#5. **a** White matter; **b** Gray matter; **c** All the segmented parts shown together

Table 6 Performance comparison of Dice score, Sensitivity and Specificity on real database

Method	Dice	Sensitivity	Specificity
Havaei et al. (2017)	85	80	84
Shreyas and Pankajakshan (2017)	83	89	86
Pereira et al. (2016)	84	86	85
Kwon et al. (2014)	86	86	86
DeepMedic (Roth et al. 2018)	86	85	86
3D ResNet (Sun et al. 2019)	85	86	86
3D CNN (Kamnitsas et al. 2017)	88	89	87
3D U-Net (Ronneberger et al. 2015)	88	88	87
Ours	**91**	**89**	**88**

Bold is indicate the optimal performance in each comparison experiment

Table 5 Computation time averaged from 100 runs of our method and 3D CNN

Method	Time (s)	
	IBSR database	BrainWeb database
3D CNN (without the speedup strategy)	340.17	755.01
Ours (without the speedup strategy)	401.31	715.05
Ours (with the speedup strategy)	**18.171**	**24.223**

Bold is indicate the optimal performance in each comparison experiment

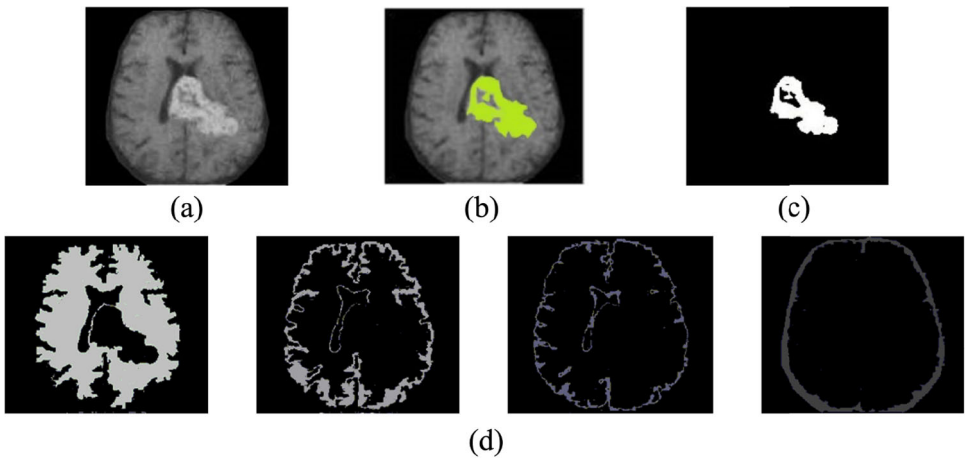


Fig. 14 Brain image segmentation and tumor extraction. **a** Original image; **b** ground truth; **c** extracted tumor; **d** segmentation result of other parts

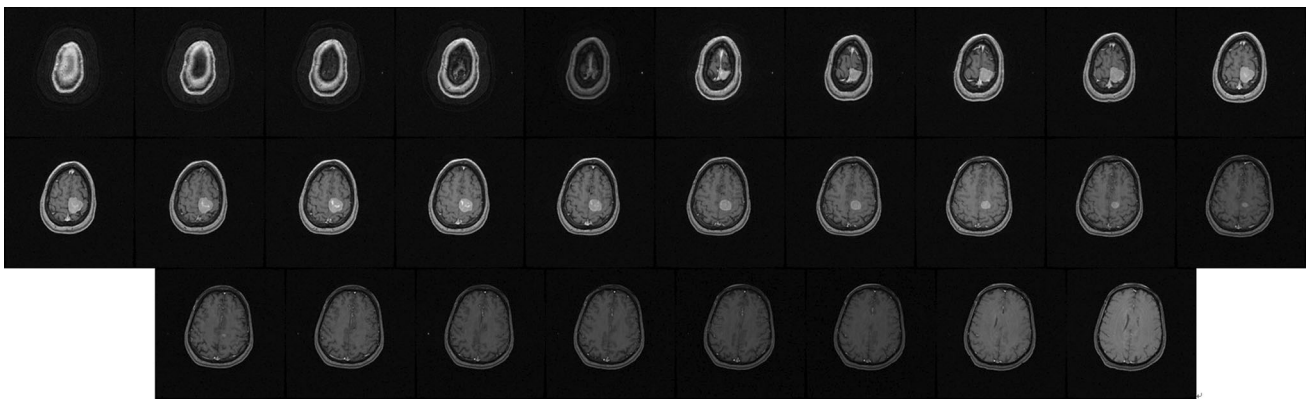


Fig. 15 Original MRI human brain volume image of sample #1

parameters for each dataset to achieve the best performance, adjusting them within a feasible range and selecting the optimal settings. As a result, we use the same architecture for simulating tests on IBSR and BrainWeb databases, while employing a different architecture for real data tests on real brain database. The top layer of the network, which has 100 units, is divided into 10 groups of 10 units each, and adjacent hidden codes are grouped together and kept at a constant size s. Note that there is no intersection between groups. When partitioning the dataset into $K_C \cdot s$ clusters, we obtain $K_C \cdot s$ hidden codes in the network. Detailed settings for all datasets and corresponding computational complexity are listed in Table 8.

Discussion

In this study, we present a VQ-based 3D segmentation model that employs a novel unsupervised 3D-DEC network and a memory reserving-and-fading strategy for efficiency enhancement. First, the validation and robustness of the proposed model are evaluated on simulated brain images from the authoritative public databases of IBSR and BrainWeb. Then, the validated model is applied to real brain MRI dataset (with tumors) for clinical practice significance. The diagrammatic expressions of the results indicate that the outcomes of our proposition have fulfilled our expectations on simulated databases (Figs. 10 and 13), compared with several improved clustering methods and 3D CNN-based models. Further comparisons with state-of-the-art algorithms on the real brain database also revealed better tumor segmentation using our proposed model (Figs. 14 and 17).

Segmentation tests on public databases of IBSR and BrainWeb

The simulated brain data used in experiments were from IBSR and BrainWeb. Previous brain segmentation studies applying various models in therapeutic areas reported remarkable success, typically clustering-related methods (Krinidis and Chatzis 2010; Kannan et al. 2012; Wang et al. 2013). In recent years, several studies have explored brain segmentation and achieved outstanding outcomes, especially since the introduction of deep learning networks (Kamnitsas et al. 2017; Sun et al. 2019; Ronneberger et al.

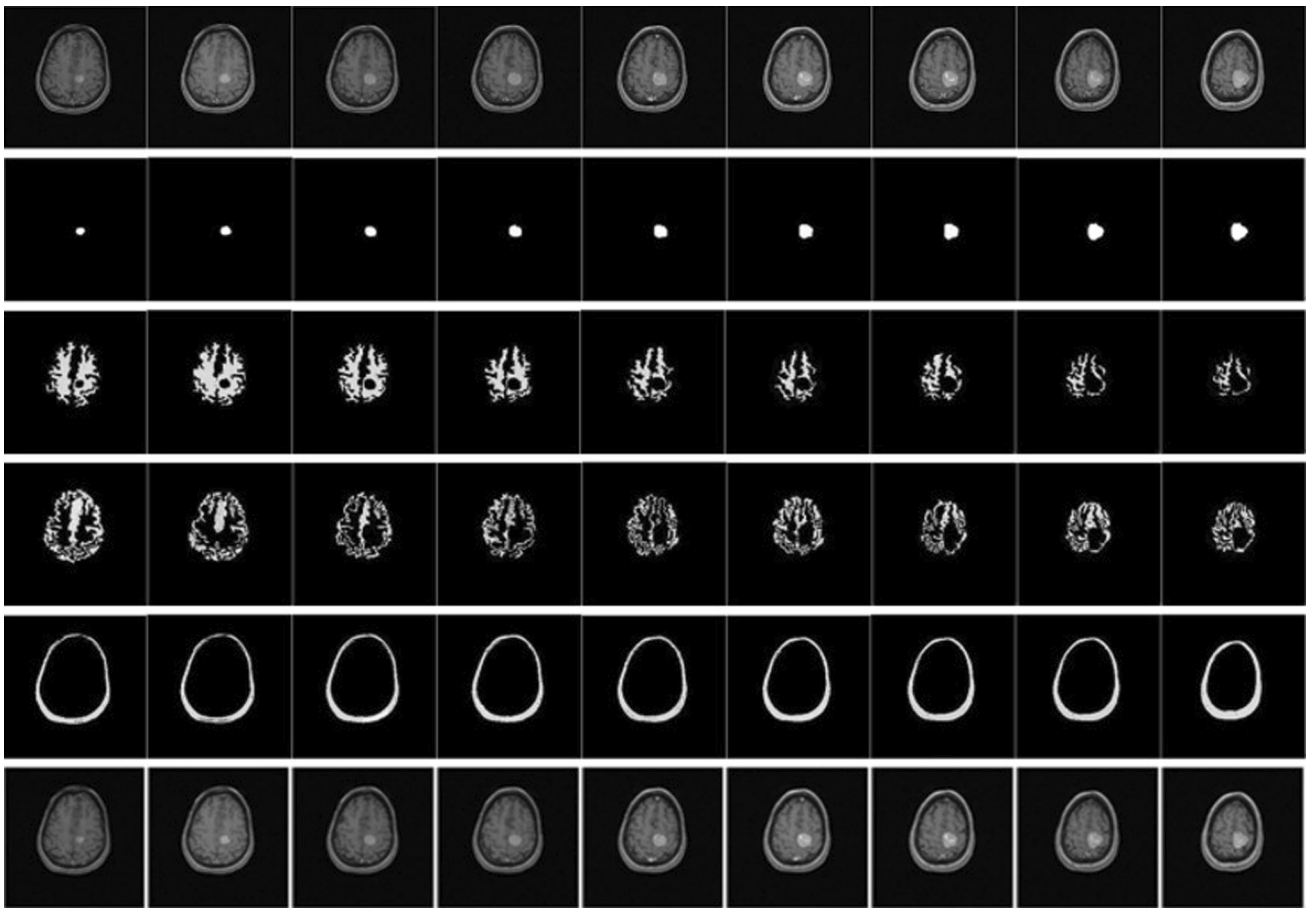


Fig. 16 Segmentation result of #1 using the proposed method. Typical sections 10, 11, 12, 13, 14, 15, 16, 17 and 18 are shown in columns from left to right. The original image, tumor findings, white matter, gray matter, skull, and ground truth are shown from top to bottom in rows

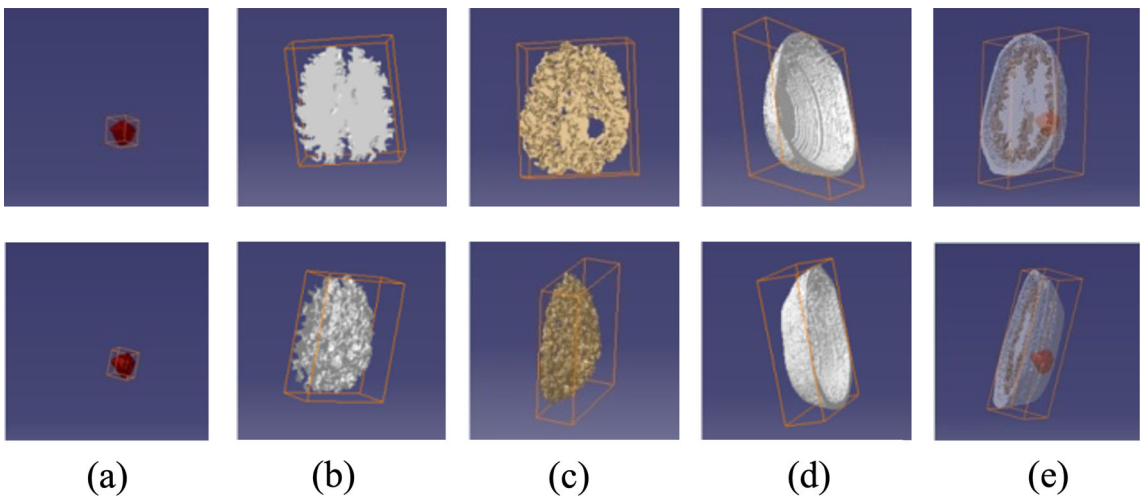


Fig. 17 Segmentation result of real 3D data. **a** Tumor result; **b** white matter result; **c** gray matter result; **d** skull result; **e** all the segmented parts shown together

2015). Our comparison studies were operated against these improved clustering methods and two 3D CNN-based methods. In comparison, 3D CNN-based models outperformed the clustering-related methods, and both CNN-based and U-Net-based models revealed excellent results for brain segmentation. For example, a recent study

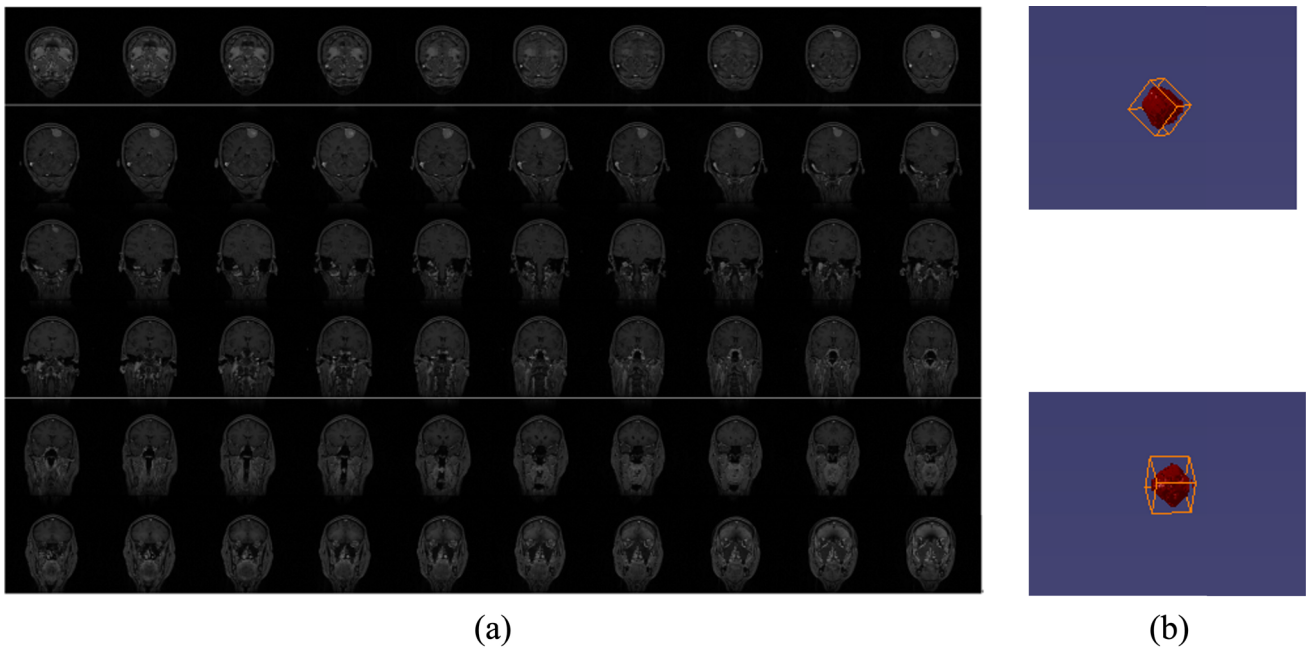


Fig. 18 Former Sample#0. **a** Original MRI image; **b** segmented tumor

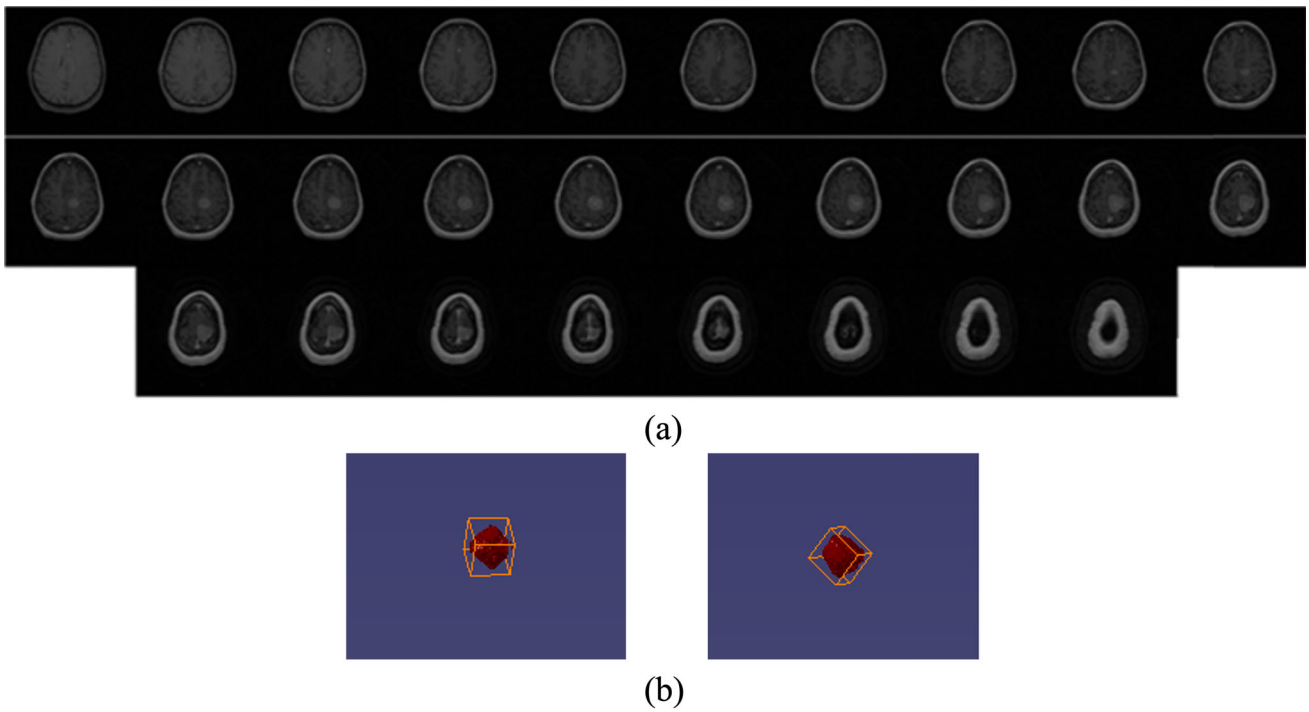


Fig. 19 Former Sample#2. **a** Original MRI image; **b** segmented tumor

using a 3D U-Net architecture to segment brain components achieved remarkable results, with a Tanimoto coefficient of 0.74 ± 0.04 (white matter segmentation) on IBSR, and with 97.4% TP and 97.7% TN (white matter segmentation) on BrainWeb (Ronneberger et al. 2015). Another study reported the application of a multi-scale 3D CNN with fully connected CRF algorithm for brain segmentation, and achieved the results of 0.73 ± 0.04 Tanimoto coefficient (white matter segmentation) on IBSR, and 97.4% TP and 97.6% TN (white matter segmentation) on BrainWeb (Kamnitsas et al. 2017). Meanwhile, our method achieved 0.74 ± 0.04 Tanimoto coefficient (white matter

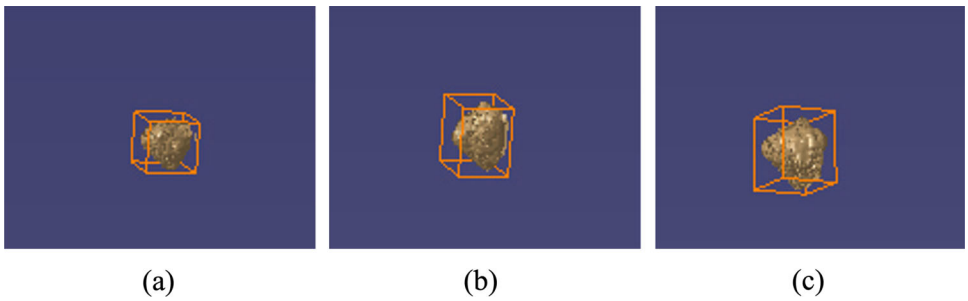


Fig. 20 Tumor development of the same patient at different periods. **a** Sample#0; **b** Sample#1; **c** Sample#2

Table 7 Ablation study of the memory reserving-and-fading strategy on real database

Method	Dice	Sensitivity	Specificity	Time (s)
3D CNN model (without speedup strategy)	89	88	88	655.03
Proposed model (without speedup strategy)	90	89	88	640.13
Proposed model (with speedup strategy)	**90**	**88**	**88**	**21.124**

Bold is indicate the optimal performance in each comparison experiment

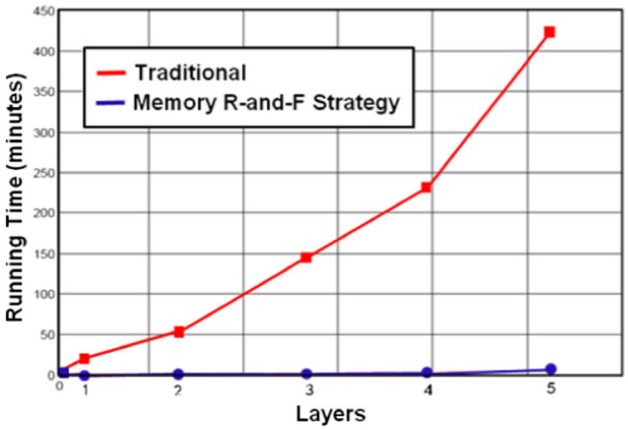


Fig. 21 Efficiency performance comparison on real database between deep network with and without a memory reserving-and-fading strategy

segmentation) on IBSR, and 97.5% TP and 97.7% TN (white matter segmentation) on BrainWeb (Table 1 and Table 3). For false instances, our model achieved a remarkable balance between recall ($\frac{TP}{TP+FN}$) and precision ($\frac{TP}{TP+FP}$), as evidenced by the results from the experiments on BrainWeb. The false negative rate (FN) was 2.5% and the false positive rate (FP) was 2.3% for white matter segmentation. This performance was superior to both the 3D CNN-based model (2.6% FN and 2.4% FP) and the 3D U-Net-based model (2.6% FN and 2.3% FP) in terms of risk and cost control.

Differ from the popular CNN-derived deep learning models nowadays, the deep learning architecture we proposed is based on VQ technology, and upgraded from the foundational DAE architecture by introducing two constraints on learned representations. One is a locality persevering constraint for embedding the original data into its underlying manifold space, and the other is a group sparsity constraint for learning block diagonal representations with non-zero groups corresponding to its clusters. The upgraded network with two new constraints learned the codebook more accurate for clustering. Therefore, superior outcomes were achieved from our segmentation work. Importantly, a well-annotated dataset is critical for accurate segmentation, since 3D CNN-based deep learning models are trained under supervision. By contrast, our VQ-based deep learning model trained the codebook in an unsupervised manner, and achieved accurate segmentation independently.

Furthermore, the computational cost of a complex and time-consuming deep learning network like 3D CNN/U-Net renders it difficult for clinical practice. Wang et al. (2021) developed a INet model to save calculation power by gradually increasing kernel sizes of convolutional

Table 8 Parameter settings and computational complexity for the proposed model

Database	Architecture	Number of parameters (M)	Time (s)	K_C	s	α	β
IBSR database	1000-500-100	6.1	18.171	5	4	0.25	0.5
BrainWeb database	1000-500-100	5.7	24.223	5	4	0.25	0.5
Real brain database	1024-680-100	9.1	21.124	6	5	0.5	1

layers, and the results turned out that 16.9% of the parameters were saved (37.6% of the parameters were saved if the model was densely connected) compared to the U-Net. As shown in Table 5, the potential speed superiority of our model was confirmed and demonstrated. For a test on IBSR, the proposed deep network-based algorithm with training strategy only took 18.171 s, while the 3D CNN model without the speedup strategy took 340.17 s, in which the total speedup ratio was 18.72. For another test on BrainWeb, our algorithm took 24.223 s while the 3D CNN model took 755.01 s, in which the total speedup ratio was 31.16. The experiments illustrated the computational efficiency of our model with/without the memory reserving-and-fading strategy. In terms of the VQ-based approach of our segmentation model, the developed speedup strategy focused on passing the learned codebook C of the "typical slice" to neighbouring slices as an initiation on the grounds of the correlation between slices, and different normalization methods were employed accordingly. Therefore, the training time of the deep learning network was reduced tremendously.

To sum up, the experiments on simulated brain databases confirmed that the proposed 3D-DEC model could gain competitive accuracy with other state-of-the-art models without depending on high-quality manually labeled data, while presenting substantial improvement in computational efficiency. It should be emphasized that the greater the increase in volume capacity, the more remarkable the improvement.

Operations on real brain MRI database and further clinical application

As described above, due to the potential performance of CNN-based and U-Net-based models for brain segmentation, the 3D U-Net architecture was also applied to real brain MRI data (Ronneberger et al. 2015). As a result, 88% Dice, 88% Sensitivity and 87% Specificity (with tumor) were revealed. A multi-scale 3D CNN with fully connected CRF algorithm to brain segmentation achieved a result of 88% Dice, 89% sensitivity and 87% specificity (with tumor) (Kamnitsas et al. 2017). By contrast, our segmentation model achieved a result of 91% Dice, 89% sensitivity and 88% specificity (with tumor), as shown in Table 6. Our model performed superior over other deep learning-related models, especially in Dice score. The outcomes were consistent with the experiments performed on simulated brain data, and the segmentation accuracy and efficiency enhancements of our model was again verified on real brain data. The proposed 3D-DEC-based model with training strategy only took 21.124 s, while the 3D CNN model (without speedup strategy) took 655.03 s, with a total speedup ratio was 31.00 (Table 7). Similarly, the proposed model without speedup strategy took 640.13 s, indicating a speedup ratio of 30.30. This confirmed that the speedup strategy proposed in our work has a prominent enhancement on algorithm efficiency for real brain MRI dataset, which is consistent with the aforementioned tests on simulated IBSR and BrainWeb databases.

Furthermore, since our model performs more meaningfully on real tumor data than on simulated data in clinical practice, more numerical parameters for segmented tumor should be calculated and more scans of the same patient should be studied (in a certain period of time). For instance, we extracted the volume size of the segmented tumor, which can be used to monitor the variation of the tumor, and evaluate the therapeutic efficacy. Figure 20 shows the lesion extraction results of the same patient at different time periods. Sample#0 was a brain tumor of 10,588 voxels in the first scan. Sample#1 was the corresponding tumor with the size of 12,839 voxels scanned 6 months later. The tumor enlarged by 2251 voxels over the period of 6 months. The total growth rate over the period was 21%, and the monthly growth rate was 3.5%. For sample#2, we extracted the same tumor of 18,346 voxels scanned 18 months later. The tumor expanded by 5507 voxels during the period of 18 months, with a total growth rate of 43% and a monthly growth rate of 2.39%. Similarly, other useful information of certain disease could be extracted through our method, providing doctors with the essential information for accurate diagnosis and precise treatment.

Ablation study

In this paper, we also conducted the ablation study on the proposed model based on the memory reserving-and-fading strategy. Both simulated brain images of public databases and real brain MRI images were segmented to evaluate the accuracy and efficiency of the proposed model compared to 3D CNN-related models, as well as the non-speedup-strategy version of our proposed model.

The accuracy comparisons were demonstrated using DSC, MHD, ASD values (for simulated databases) and Dice, Sensitivity, Specificity values (for real database) in experiments. As shown in Table 4 and Table 6, 3D CNN and U-Net models yielded 0.87 DSC, 6.81 MHD, 0.381 ASD, and 0.88 DSC, 6.67 MHD, 0.374 ASD respectively, while our model yielded 0.92 DSC, 6.44 MHD, 0.361 ASD for tests on simulated brain images of public databases. 3D CNN and U-Net models gave 88% Dice, 89% Sensitivity, 87% Specificity values, and 88% Dice, 88% Sensitivity, 87% Specificity values respectively, while our model gave 91% Dice, 89% Sensitivity, 88% Specificity for tests on real brain MRI database. The efficiency comparisons were illustrated using running time in experiments. As shown in Table 5 and Table 7, for simulated databases, the proposed

algorithm with training strategy had a speedup ratio of 18.72 compared to the 3D CNN model on the IBSR database, and had a speedup ratio of 31.16 compared to the 3D CNN model on the BrainWeb (discussed above). Meanwhile, the proposed algorithm without speedup strategy achieved a running time of 401.31 s on IBSR, and 715.05 s on BrainWeb, representing the total speedup ratios of 22.09 and 29.52 compared to our proposed model, respectively. For real brain MRI database, the proposed model with training strategy had a speedup ratio of 31.00 compared to the 3D CNN model (discussed above), and 30.30 compared to the same model architecture without speedup strategy with a running time of 640.13 s (Table 7).

Therefore, the ablation study validated the accuracy of the designed 3D-DEC network was validated, as well as the efficiency of the proposed memory reserving-and-fading strategy. Moreover, Fig. 21 clearly demonstrates the speedup effectiveness of the memory reserving-and-fading strategy in a deep network, and its evolution with the depth of the network.

Limitations

Although our work performs well for brain segmentations in terms of accuracy and computational efficiency, there are still some limitations that need to be considered.

The simulated samples employed in our study are limited to databases of IBSR and BrainWeb, and more public databases need to be developed to further improve the generalization ability of our model. Meanwhile, the capacity of the real brain MRI database we established also need to be continuously expanded in following work.

Other than the volume parameter of extracted tumors after segmentation, more numerical results that precisely describe the lesions should be provided for clinical assistance by further analysis and customized parameter extraction.

Besides the application of brain tumor analysis in this work, the extraction and segmentation work for other types of lesions should be discussed in the future.

Remarks

- Our study aims to generalize the promising theoretical research to useful clinical applications. The proposed segmentation model is performed both on simulated data and real data with lesions. After lots efforts engaged, a real brain MRI image database with tumors has been established in our institution, and the capacity of which is constantly growing.
- The motivation of our work is to overcome the obstacles found in practical applications. For example, the proposed model overcomes the lack of high-quality expert-labelled data in commonly used methods nowadays. We also propose an appropriate solution to enhance the model efficiency.
- Since deep learning-based neural networks become a methodology of choice for analyzing medical images, our study achieves superior outcomes in terms of segmentation accuracy evaluated by multiple criteria.
- Our method is applied to brain tumor analysis and clinical assistance, which is feasible to generalize to lesions of other diseases in medical image processing and computer-aided diagnosis.

Conclusion

In this study, the proposed method for segmentation based on the VQ technique, implemented by a 3D deep learning model and enhanced by the memory reserving-and-fading strategy, was proven to be accurate and efficient in 3D medical image processing, for both simulated samples and real samples without manual annotations. Specifically, a highly computational deep network named 3D-DEC model was embedded into the system, realizing the key process of VQ in an unsupervised manner. Meanwhile, a novel memory reserving-and-fading strategy was brought into the model training for the purpose of efficiency enhancement.

Through extensive experiments, first, we found that the proposed model achieved better precision than other state-of-the-art deep networks-based models. Second, by introducing our acceleration strategy, the efficiency of the proposed deep learning network training was enormously improved, and this advantage had a greater impact on real patients with brain tumors.

Furthermore, our proposed method can be used to assist in radiotherapy planning, postoperative follow-up and target delineation, and has great potential in processing other type of medical images, such as lung CT images and liver MRI images. With further efforts in the future, the proposed model is extendible for more clinical applications other than segmentation, such as prediction and diagnosis.

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